Business Requirements Document (BRD)

Smart Healthcare Data Analysis & Power BI Dashboard Build

Signature:

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1. Business Understanding

The attached dataset represents operational records from a "Smart Healthcare" system, capturing patient data across multiple hospitals/branches spread over 10 Egyptian cities (Cairo, Giza, Alexandria, Luxor, Aswan, Tanta, Port Said, Suez, Ismailia, Mansoura), and covering 15 medical departments (Emergency, Internal Medicine, General Surgery, Pediatrics, Cardiology, Oncology, Dermatology, Orthopedics, Neurology, ICU, etc.).

Each record represents one patient's full treatment journey from Admission to Discharge, and includes: patient demographics (age, gender, blood type, city), visit details (admission/discharge dates, department, attending doctor, diagnosis, symptoms, treatment, lab status), financial data (insurance type, medical cost), and finally performance/quality indicators (length of stay, readmission, patient outcome, satisfaction score).

The end goal is to turn this raw data (with its quality issues) into a reliable Power BI dashboard that supports decision-making at three levels:

- Operational: tracking patient volume, busiest departments, average length of stay.
- Financial: treatment cost by department and insurance type, cost optimization.
- Clinical Quality: readmission rates, patient outcomes, patient satisfaction.

1.1 Stakeholders

Stakeholder	Primary Interest
Executive Management / Hospital Director	Overall performance, cost, and quality overview
Medical Department Heads	Patient volume, case status, and outcome rates per department
Finance Department	Treatment cost, insurance mix, cost optimization
Quality & Patient Experience Team	Patient satisfaction, readmissions, critical cases
Data Analytics / IT Team	Data cleansing and building/maintaining the dashboard

1.2 Project Scope

- Clean and standardize the Smart_Healthcare_Dirty_Data_800K.csv dataset (800,000 records).
- Build a reliable clean data model suitable as a Power BI source.
- Design a two-page dashboard (Executive/Operational + Financial/Clinical Quality).
- Out of scope: real-time system integration or predictive modeling at this stage.

2. Data Quality Assessment & Business Impact

The dataset (800,000 rows / 20 columns) was profiled and several data quality issues were found that prevent using the data directly for the dashboard without cleansing. The tables below summarize each issue, its scale, business impact, and the proposed fix.

2.1 Missing Values

Column	Missing Count	Business Impact	Proposed Solution
Satisfaction_Score	150,157 (~18.8%)	Distorts any patient satisfaction KPI and reduces its reliability	Exclude NULLs from the average calculation; add a card showing the % of unrated visits

Column	Missing Count	Business Impact	Proposed Solution
Symptoms	93,077 (~11.6%)	Weakens analysis of common symptoms per diagnosis	Tag as "Not Recorded" instead of removing; keep out of primary Page 1 visuals
Lab_Status	15,959 (~2%)	Affects analysis of critical cases in Emergency	Tag as "Unspecified" and show as its own category in the chart
Insurance_Type	12,018 (~1.5%)	Reduces accuracy of cost-by-insurance analysis	Tag as "Unspecified" instead of dropping the record
City	7,974 (~1%)	Loses some records from the geographic breakdown	Tag as "Unspecified"; keep the record (rest of its data is valid)
Patient_Outcome	7,985 (~1%)	Reduces accuracy of recovery/mortality rates	Exclude only from the outcome-rate denominator, not from other metrics
Doctor	7,942 (~1%)	Weakens per-doctor performance analysis	Tag as "Unspecified"
Treatment	8,834 (~1.1%)	Reduces accuracy of treatment-type analysis	Tag as "Unspecified"
Diagnosis	7,268 (~0.9%)	Reduces accuracy of top-diagnosis analysis	Tag as "Unspecified"
Discharge_Date	4,919 (~0.6%)	Prevents computing length of stay for these rows (LOS column already exists though)	Rely on the existing Length_of_Stay_Days field instead of recalculating from dates
Department	971 (~0.1%)	Loses some records from department analysis	Tag as "Unspecified"
Medical_Cost	999 (~0.1%)	Reduces accuracy of total cost	Exclude only from cost totals/averages
Age	1,085 (~0.1%)	Reduces accuracy of age-band analysis	Exclude only from age-band analysis
Readmission	462 (~0.06%)	Minor effect on the readmission rate	Tag as "Unspecified"

2.2 Inconsistent Formatting

Column	Issue Found	Business Impact	Proposed Solution
Gender	9 variants of the same value: Male / male / " Female" / female / M / F / "Male " / "?" (450 rows)	Same category splits across multiple chart slices instead of just 2 (Male/Female)	Trim + standardize casing, map abbreviations (M→Male, F→Female); classify "?" as "Unspecified"
Department	Repeated in different forms: Emergency/emergency/ER, Cardiology/cardiology/Cardio	Same department's data is split across several categories, understating its real volume	Standardize names via a mapping table for every abbreviation to the official department name

Column	Issue Found	Business Impact	Proposed Solution
Readmission	8 variants: Yes/No/1/0/yes/N/NO/Y instead of just 2 values	Readmission rate (a core quality KPI) is calculated incorrectly	Standardize all variants to Yes / No before any calculation
Medical_Cost	Stored as text, not a number: contains a $ sign, thousands separators (,), and a stray character (u) in 479 rows	Cost cannot be summed or averaged directly in Power BI	Strip the symbols and convert the column to a numeric (Decimal) type during data prep (Power Query)
Symptoms	Multiple symptoms sometimes packed into one cell, separated by ";"	Makes it hard to analyze how often each individual symptom occurs	Split/unpivot the multi-value column when a per-symptom breakdown is needed

2.3 Invalid / Out-of-Range Values

Column	Issue Found	Business Impact	Proposed Solution
Age	554 rows with negative age, and 532 rows with age above 110 (up to 150)	Distorts average age and age-band analysis, producing clinically nonsensical results	Exclude values outside the valid range (0–110) from age analysis and tag as "Invalid"
Medical_Cost	575 rows with negative cost (down to -500)	Negative cost is not financially meaningful and understates the true cost total	Exclude negative values from cost totals/averages; flag them for review as data-entry errors
Doctor	Only 8 doctors on record for 800,000 visits (~99K cases each)	Not operationally realistic; suggests the data is partly synthetic/generated rather than fully real	Disclose to decision-makers that "doctor performance" analysis in this version is indicative only, and request real doctor data from the source system (HIS) going forward

2.4 Data Strengths

- No fully duplicated rows, and no duplicate Patient_ID values (every patient has a unique ID).
- Admission and discharge dates are correctly formatted, and no discharge date precedes its admission date.
- Length_of_Stay_Days is already numeric with no negative values, and can be used directly.
- Blood_Type is fully standardized with no formatting issues (exactly 8 blood types, as clinically expected).

2.5 Data Privacy Note

The Patient_Name column contains personally identifiable information (PII) that should not be displayed in a management/analytics dashboard. It is recommended to exclude it from all dashboard visuals and rely on Patient_ID for individual patient lookups when needed, with access to that level of detail restricted appropriately.

3. Recommended Data Preparation Plan

Before building the dashboard, the data must go through a preparation stage (Power Query inside Power BI, or a separate ETL step) that includes the following:

- Convert numeric columns (Age, Medical_Cost) from text to numbers after stripping symbols and separators.
- Standardize categorical text columns (Gender, Department, Readmission) using mapping/lookup tables.
- Apply validation rules: Age between 0 and 110, Cost ≥ 0; otherwise exclude the value from calculations while keeping the record.
- Add an "Unspecified" value for blank entries in categorical columns instead of leaving them empty (so they appear as a clear category in filters instead of disappearing).
- Build a Date Table from Admission_Date to support time-trend analysis (monthly/quarterly).
- Exclude Patient_Name from the data model used in the dashboard for privacy reasons.

4. Power BI Dashboard Design

It is recommended to build the dashboard as two integrated pages, plus a set of global slicers fixed at the top of each page.

4.1 Global Slicers — appear on both pages

- Date range (Admission_Date)
- City
- Department
- Insurance_Type
- Gender

4.2 Page 1: Operational Overview

Goal: give management and department heads a quick view of patient volume, department activity, and service quality.

Cards / KPIs:

Card	Metric / Calculation
Total Patients	COUNT(Patient_ID)
Average Length of Stay (days)	AVERAGE(Length_of_Stay_Days)
Readmission Rate	COUNT(Readmission=Yes) / COUNT(Total), after standardizing values
Average Treatment Cost	AVERAGE(Medical_Cost), after excluding invalid values
Average Patient Satisfaction Score	AVERAGE(Satisfaction_Score), after excluding blanks
% Critical Cases	COUNT(Lab_Status=Critical) / COUNT(Total)

Charts:

Chart	Type	Purpose
Patients by Department	Bar Chart	Identify the busiest departments by volume
Patients by City	Map / Bar Chart	Geographic distribution of cases

Chart	Type	Purpose
Monthly Patient Trend	Line Chart	Track patient volume growth/decline over time
Patient Outcome Distribution	Donut Chart	Share of Recovered / Stable / Deceased, etc.
Lab Status Distribution	Bar Chart	Share of Critical / High / Normal cases
Average Length of Stay by Department	Bar Chart	Compare department efficiency in treatment speed

4.3 Page 2: Financial & Clinical Quality

Goal: support financial decision-making and measure the quality of care delivered.

Cards / KPIs:

Card	Metric / Calculation
Total Treatment Cost	SUM(Medical_Cost), after excluding negative/blank values
Average Cost per Patient	AVERAGE(Medical_Cost)
Mortality Rate	COUNT(Patient_Outcome=Deceased) / COUNT(Total)
% Left Against Medical Advice	COUNT(Patient_Outcome=Left Against Advice) / COUNT(Total)
Highest-Cost Department	MAX(SUM(Medical_Cost) by Department)
% Patients Without a Satisfaction Rating	COUNT(Satisfaction_Score=NULL) / COUNT(Total)

Charts:

Chart	Type	Purpose
Total Cost by Department	Bar Chart	Identify the highest-cost departments for cost optimization
Cost by Insurance Type	Donut / Bar Chart	Understand the financial burden split by payer
Length of Stay vs. Cost	Scatter Chart	Spot cases with unusually high cost relative to their stay duration
Readmission Rate by Department	Bar Chart	Identify departments needing a quality review
Average Satisfaction by Department	Bar Chart	Measure patient experience per department
Top 10 Diagnoses by Case Count	Bar Chart (Top N)	Focus on the most common conditions for clinical planning

5. Success Criteria

- Categorical columns used in the dashboard (Gender, Department, Readmission) are 100% clean and standardized.
- Numeric columns (Age, Medical_Cost) are directly computable in Power BI with no data-type conversion errors.
- The dashboard loads and refreshes without errors across the full 800,000 records.
- Every card and chart on both pages is tied to a documented data source and calculation, as described in this document.